Math 110 — Week 4

Solutions

HW4 P1.

Idea. The identity in (a) hands you the inverse when $ad - bc \neq 0$, and forces $A = 0$ when $ad - bc = 0$ and A is assumed invertible.

1. (a) Write $B = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$.
$$AB = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)I.$$
Likewise
$$BA = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = (ad - bc)I.$$
2. (b) If $\delta = ad - bc \neq 0$, then $A(\delta^{-1}B) = I = (\delta^{-1}B)A$. So A is invertible with $A^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$.
3. (c) Suppose $ad - bc = 0$ and A is invertible. Then $AB = 0$, and multiplying on the left by A^{-1} gives $B = 0$. So $d = b = c = a = 0$, i.e. $A = 0$. But the zero matrix is not invertible. Contradiction.

HW4 P2.

Idea. Compute $ad - bc$ in the stated field.

1. $\left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$: $6 - 5 = 1$. Invertible, inverse $\left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$.
2. $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$: $(1)(-1) - i \cdot i = -1 + 1 = 0$. Not invertible. (Second column is i times the first.)
3. $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$ over $\mathbb{F}_5$: $1 - 6 = -5 = 0$ in $\mathbb{F}_5$. Not invertible. (Second column is 2 times the first: $2 \cdot 3 = 6 = 1$.)

HW4 P3.

Idea. Injective maps keep independence; n independent vectors in dimension n are a basis. Conversely a basis in the range makes T surjective.

1. (a)⇒(b). Let $v_1, \ldots, v_n$ be a basis. If $\sum \lambda_j T v_j = 0$ then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j = 0$ (T injective), so all $\lambda_j = 0$.
2. So $Tv_1, \ldots, Tv_n$ are n independent vectors in V, $\dim V = n$: a basis.
3. (b)⇒(a). Take any basis $v_1, \ldots, v_n$. Then $\operatorname{range} T \supseteq \operatorname{span}(Tv_1, \ldots, Tv_n) = V$, so T is surjective, hence an isomorphism (Prop. 31).

HW4 P4.

Idea. Rank is the dimension of the range. Compare the range of BC with the range of B, and view it as the image of $\operatorname{range} C$.

1. Every $BCx = B(Cx)$ is in $\operatorname{range} B$. So $\operatorname{range}(BC) \subseteq \operatorname{range} B$ and $\operatorname{rank}(BC) \leq \operatorname{rank} B$.
2. $\operatorname{range}(BC) = \{By : y \in \operatorname{range} C\}$ is the range of the restriction $B|_{\operatorname{range} C}$. By rank–nullity for this restriction, its dimension is $\dim \operatorname{range} C - \dim \operatorname{null}(B|_{\operatorname{range} C}) \leq \operatorname{rank} C$.
3. (b) Take $B = C = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: both have rank 1, and $BC = 0$ has rank 0.

HW4 P5.

Idea. Put each basis in its own slot.

1. Let $v_1, \dots, v_n$ and $w_1, \dots, w_m$ be bases of V and W. Claim: $(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)$ is a basis of $V \times W$.
2. Spans:
$$(v, w) = (\sum a_i v_i, \sum b_j w_j) = \sum a_i (v_i, 0) + \sum b_j (0, w_j).$$
3. Independent:
$$\sum a_i (v_i, 0) + \sum b_j (0, w_j) = (\sum a_i v_i, \sum b_j w_j) = (0, 0)$$
gives $\sum a_i v_i = 0$ and $\sum b_j w_j = 0$, so all $a_i = 0$ and all $b_j = 0$.
4. The list has $n + m$ vectors.

HW4 P6.

Idea. A subspace contains 0; that locates v.

1. (a) If $v + U$ is a subspace, $0 \in v + U$, so $0 = v + u$ and $v = -u \in U$.
2. If $v \in U$: $v + u \in U$ for all $u \in U$, so $v + U \subseteq U$; and each $u = v + (u - v) \in v + U$, so $U \subseteq v + U$. So $v + U = U$, a subspace.
3. (b) If $v + U = v' + U$, then $v = v + 0 \in v' + U$, so $v = v' + u$ and $v - v' \in U$.
4. If $v - v' \in U$: $v + u = v' + ((v - v') + u) \in v' + U$, so $v + U \subseteq v' + U$; since $v' - v \in U$ as well, the reverse inclusion holds.

HW4 P7.

Idea. An invertible operator sends a basis to a basis. Build a basis of V starting in U, and a second basis starting with the images under S.

1. Let $u_1, \dots, u_m$ be a basis of U; extend to a basis $u_1, \dots, u_m, x_1, \dots, x_k$ of V.
2. $Su_1, \dots, Su_m$ is independent since S is injective. Extend it to a basis $Su_1, \dots, Su_m, y_1, \dots, y_k$ of V (the same k, since both bases have $\dim V$ vectors).
3. Define T by $Tu_i = Su_i$, $Tx_j = y_j$. Then T and S agree on a basis of U, so $T|_U = S$.
4. T sends a basis to a basis, so it is surjective, hence invertible (Prop. 31).

R1.

Idea. S must undo T on $\operatorname{range} T$; what it does elsewhere is free. Use a basis of W that starts with $Tv_1, \dots, Tv_n$.

1. (a) If $\sum \lambda_j T v_j = 0$ then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j = 0$, so all $\lambda_j = 0$.
2. (b) Extend $Tv_1, \dots, Tv_n$ to a basis $Tv_1, \dots, Tv_n, w_1, \dots, w_k$ of W. Define S by $S(Tv_j) = v_j$ and $S(w_i) = 0$.
3. Then $ST(v_j) = v_j$ for each j: ST and I_V agree on a basis, so $ST = I_V$.

R2.

Idea. This is the map Γ.

$\Gamma(u, w) = u + w$ is linear and maps onto $U + W$. If $u + w = 0$ then $u = -w \in U \cap W = \{0\}$, so $u = w = 0$: injective. So $\Gamma \colon U \times W \to U \oplus W$ is an isomorphism.

R3.

Idea. Read off null space and range.

(i) If $R(a) = 0$ then every $a_j = 0$: injective. The sequence $(1, 0, 0, \dots)$ is not in the range (every output starts with 0): not surjective. (ii) For an operator on a finite-dimensional space, injective implies surjective (Prop. 31).

R4.

Idea. Split v as $(v - Pv) + Pv$.

1. $Pv \in \operatorname{range} P$, and $P(v - Pv) = Pv - P^2v = 0$, so $v - Pv \in \operatorname{null} P$. Hence $V = \operatorname{null} P + \operatorname{range} P$.
2. If $v \in \operatorname{null} P \cap \operatorname{range} P$, write $v = Pw$. Then $v = Pw = P^2w = Pv = 0$. So the sum is direct.

R5.

Idea. The matrix says $D\beta_1 = \gamma_1$, $D\beta_2 = \gamma_2$, $D\beta_3 = \gamma_3$, $D\beta_4 = 0$. Put the null space last and let γ be the images.

$\beta = (x,\ x^2,\ x^3,\ 1)$ and $\gamma = (1,\ 2x,\ 3x^2)$. Both are bases (distinct degrees), and $Dx = 1$, $Dx^2 = 2x$, $Dx^3 = 3x^2$, $D1 = 0$.